DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech. **Semester :VII**

Branch: Electronics & Telecommunication Engineering /Electronics and Communication Engineering
Subject Code & Name: BTETC701 Microwave Engineering

Max Marks: 60 **Date:05/02/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	The modes of propagation supported by a rectangular wave guide is:				CO1	1
	a. TM, TEM, TE	b. TM, TE	c. TM, TEM	d. TE, TEM		
2	The Smith chart is based on the polar plot of				CO1	1
	a. Voltage reflection coefficient	b. Impedance	c. Reactance	d. Current		
3	A cylindrical cavity resonator can be constructed using a circular waveguide.				CO1	1
	a. shorted at both the ends	b. open at both the ends	c. matched at both the ends	d. none of the mentioned		
4	The device used to get the measurement of S parameters of n- port micro wave network is				CO2	1
	a. CRO	b. Network analyzer	c. Attenuator	d. Circulator		
5	Ferrite isolators are _____ port microwave devices.				CO2	1
	a. Eight	b. Three	c. Two	d. Four		
6	Which device is based on Faraday rotation?				CO2	1
	a. Magic Tee	b. E plane Tee	c. Isolator	d. H plane tee		
7	_________ is a single cavity klystron tube that operates as on oscillator by using a reflector electrode after the cavity.				CO3	1
	a. Backward wave oscillator	b. Reflex klystron	c. Travelling wave tube	d. Magnetrons		
8	Klystron operates on the principle of				CO3	1
	a. Amplitude Modulation	b. Frequency Modulation	c. Pulse Modulation	d. Velocity Modulation		
9	The range of VSWR in dB indicated in the VSWR meter is _________				CO4	1
	a. 0–2 dB	b. 0–10 dB	c. 0–5 dB	d. 0–100 dB		
10	The Double Minimum method is relevant to the measurement of				CO4	1

	a. low attenuation coefficient	b. high attenuation coefficient	c. high VSWR	d. low VSWR		
11	Stripline can be compared to a:				CO5	1
	a. Flattened rectangular waveguide	b. Flattened circular waveguide	c. Flattened co axial cable	d. None of the mentioned		
12	The mode of propagation in a microstrip line is				CO5	1
	a. Quasi TEM	b. TEM	c. TM	d. TE		
Q. 2	**Solve the following.**					**12**
A)	An air filled rectangular waveguide of inside dimension 7 X 3.5 cm, operates in dominant TE_{10} mode. Find 1. Cut off Frequency 2. Determine the phase velocity of wave in guide at 3.5 GHz. 3. Determine the guided wavelength at same frequency.				CO1	**6**
B)	Compare transmission lines and waveguides.				CO1	**6**
Q.3	**Solve the following.**					**12**
A)	Explain E- plane Tee with help of S matrix.				CO2	**6**
B)	Explain working principal of directional coupler with help of neat and labeled diagram. State and explain its performance parameters.				CO2	**6**
Q. 4	**Solve Any Two of the following.**					**12**
A)	Explain Construction and Principle of operation of helix type travelling wave tube (TWT) in detail.				CO3	**6**
B)	Explain Reflex klystron with help of Applegate diagram.				CO3	**6**
C)	Explain Construction and Principle of operation of 8 cavity cylindrical travelling wave magnetron.				CO3	**6**
Q.5	**Solve Any Two of the following.**					**12**
A)	Explain Impedance measurement techniques at microwave frequency.				CO4	**6**
B)	Define VSWR. Explain the VSWR meter with the help of diagram.				CO4	**6**
C)	Explain the power measurement technique with the help of diagram.				CO4	**6**
Q. 6	**Solve Any Two of the following.**					**12**
A)	Explain the micro strip lines in detail with neat diagram. State its applications.				CO5	**6**
B)	Explain Structural details and applications of Stripline.				CO5	**6**
C)	What are the hazards of Electromagnetic Radiation?				CO5	**6**
	***** End *****					